

SUPERSTORE SALES ANALYSIS

AN END TO END DATA ANALYTICS PROJECT

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ABSTRACT

This project presents an end to end sales analysis of the Superstore dataset using **Microsoft Excel, PostgreSQL, and Microsoft Power BI**. The objective was to transform raw sales data into meaningful business insights through data cleaning, exploratory data analysis, and interactive dashboard development. Microsoft Excel was used to clean and prepare the dataset by removing duplicates, checking for missing values, formatting dates, and creating additional analytical columns. The cleaned data was then imported into PostgreSQL, where SQL queries were used to explore business performance and calculate key metrics such as total sales, profit, order volume, and category performance. Finally, Power BI was used to create an interactive dashboard featuring KPIs, charts, and slicers that allow users to explore the data dynamically. The findings provide insights into sales trends, regional profitability, customer segments, and top performing products, demonstrating the value of business intelligence in supporting data driven decision making.

Tools Used:

Microsoft Excel • PostgreSQL • Microsoft Power BI

1. INTRODUCTION

In today's business environment, organizations generate vast amounts of data from their daily operations, including sales transactions, customer purchases, inventory records, and financial activities. While this data is valuable, it has little practical use unless it is properly analyzed and transformed into meaningful information. Business Intelligence (BI) plays a crucial role in this process by enabling organizations to collect, analyze, visualize, and interpret data to support informed decision making.

Business Intelligence refers to the technologies, tools, and processes used to convert raw data into actionable insights. Rather than relying on assumptions or intuition, businesses use BI to monitor performance, identify trends, evaluate customer behavior, measure profitability, and make strategic decisions based on accurate and timely information. Modern organizations across industries including retail, healthcare, finance, manufacturing, and education use Business Intelligence to improve efficiency, optimize operations, and gain a competitive advantage.

One of the most common applications of Business Intelligence is the analysis of sales data. Companies analyze sales data to understand how products perform, identify their most profitable customer segments, monitor regional performance, evaluate seasonal trends, and measure overall business growth. Sales analysis enables managers to identify strengths and weaknesses within the business, allocate resources effectively, improve customer satisfaction, and develop strategies that increase revenue and profitability.

Microsoft Power BI has become one of the leading Business Intelligence tools because of its ability to transform complex datasets into interactive dashboards and reports. It allows users to connect data from multiple sources, perform data modeling, create calculations using Data Analysis Expressions (DAX), and develop dynamic visualizations that support real time analysis. Features such as KPI cards, charts, maps, slicers, and drill down capabilities make Power BI an effective tool for communicating business performance to both technical and non technical stakeholders.

This project demonstrates a complete end to end data analytics workflow using the **Superstore Sales Dataset**. The project began with data cleaning and preparation in **Microsoft Excel**, where

duplicate records were removed, missing values were checked, dates were standardized, and additional analytical columns were created. The cleaned dataset was then imported into **PostgreSQL**, where SQL queries were used to perform exploratory data analysis (EDA), calculate key business metrics, and validate the dataset. Finally, the processed data was connected to **Microsoft Power BI**, where DAX measures, KPI cards, interactive charts, and slicers were developed to create a comprehensive sales performance dashboard.

The resulting dashboard provides an overview of key business metrics, including total sales, total profit, total orders, average order value, monthly sales trends, regional profitability, customer segment distribution, product category performance, and the top performing products. Interactive filters enable users to analyze performance by **Year, Month, Region, Category, and Customer Segment**, making the dashboard flexible and suitable for different business reporting needs.

The primary purpose of this project is to demonstrate the complete data analytics process—from raw data preparation to business intelligence reporting—while showcasing practical skills in Microsoft Excel, PostgreSQL, Power BI, data visualization, and business insight generation. By completing this project, I developed the ability to clean and prepare data, perform SQL based analysis, build interactive dashboards, and communicate findings through clear visualizations and actionable recommendations. This project serves as both a demonstration of technical proficiency and a practical example of how Business Intelligence can support data driven decision making in a retail business environment.

PROJECT OBJECTIVES

The primary objective of this project was to perform an end to end sales data analysis using Microsoft Excel, PostgreSQL and Microsoft Power BI. The project aimed to transform raw sales data into meaningful insights that support data driven decision making while demonstrating the complete data analytics workflow.

The specific objectives of the project were to:

- Clean and prepare the raw Superstore sales dataset in Microsoft Excel by removing duplicate records, checking for missing values, standardizing date formats, and creating additional columns required for analysis.

- Improve the quality and reliability of the dataset to ensure accurate analysis and reporting.
- Import the cleaned dataset into PostgreSQL and organize it within a relational database for efficient storage and management.
- Perform Exploratory Data Analysis (EDA) using SQL queries to understand the dataset, identify trends, and calculate important business metrics such as total sales, total profit, total orders, average order value, product performance, and regional profitability.
- Validate the accuracy of the imported data before proceeding to visualization and reporting.
- Connect PostgreSQL to Microsoft Power BI to create a reliable data source for dashboard development.
- Develop DAX measures to calculate key performance indicators (KPIs), including Total Sales, Total Profit, Total Orders, and Average Order Value.
- Design an interactive Power BI dashboard that presents business performance through KPI cards, charts, and slicers, allowing users to explore data from multiple perspectives.
- Analyze sales performance across different dimensions, including time, product category, customer segment, region, and individual products.
- Generate meaningful business insights from the dashboard to identify sales trends, customer behavior, and high performing products and regions.
- Provide practical recommendations that can help improve sales performance, profitability, and business decision making.
- Demonstrate practical proficiency in Microsoft Excel, PostgreSQL, SQL, and Microsoft Power BI by completing an end to end data analytics project suitable for a professional portfolio.

These objectives guided every stage of the project, from data preparation and database management to dashboard development and business insight generation, ensuring that the final output was both technically sound and valuable for business analysis.

2. DATASET OVERVIEW

2.1 Dataset Description

This project utilized the **Superstore Sales Dataset**, a widely used retail dataset commonly employed for learning data analytics, SQL, and business intelligence concepts. The dataset contains historical sales transactions from a retail business and provides detailed information about customers, products, sales performance, profitability, shipping, and geographical locations.

The dataset was selected because it represents a realistic business environment and contains sufficient information to perform exploratory data analysis (EDA), sales performance evaluation, customer segmentation, regional analysis, and dashboard development.

Before analysis, the dataset underwent a comprehensive data preparation process in Microsoft Excel to ensure that it was clean, consistent, and suitable for further analysis in PostgreSQL and Microsoft Power BI.

2.2 Number of Records

The cleaned dataset used in this project contains **9,994 sales transactions**, representing individual customer orders made over a four year period.

Each record corresponds to a unique sales transaction and includes information about the customer, purchased product, order date, sales value, profit, discount, quantity purchased, shipping details, and geographical location.

2.3 Key Variables in the Dataset

The dataset contains several variables that were used throughout the analysis. The most important variables are summarized below.

Column	Description
Order ID	Unique identifier assigned to each customer order.

Column	Description
Order Date	Date on which the order was placed.
Ship Date	Date the order was shipped to the customer.
Ship Mode	Shipping method used for delivery.
Customer ID	Unique identifier for each customer.
Customer Name	Name of the customer who placed the order.
Segment	Customer category (Consumer, Corporate, or Home Office).
Country	Country where the customer is located.
City	Customer's city.
State	Customer's state or region.
Region	Geographic sales region (Central, East, South, and West).
Product ID	Unique identifier assigned to each product.
Category	Product category (Technology, Furniture, and Office Supplies).
Sub Category	Specific classification of the product within each category.
Product Name	Name of the purchased product.
Sales	Revenue generated from each transaction.
Quantity	Number of units purchased.
Discount	Discount applied to the transaction.
Profit	Profit earned from each transaction.

2.4 Importance of the Dataset

The Superstore dataset provides sufficient information to evaluate different aspects of business performance. By analyzing the available variables, it is possible to:

- Measure the company's total sales and profit.
- Monitor monthly and yearly sales trends.
- Compare product category performance.
- Identify the most profitable regions.
- Analyze customer purchasing behavior across different segments.
- Determine the best selling products.
- Evaluate order volumes and average order values.
- Generate business insights that support strategic decision making.

These variables formed the foundation for the SQL analysis and the interactive Power BI dashboard developed during this project.

2.5 Data Preparation Summary

Before importing the dataset into PostgreSQL, several preprocessing tasks were carried out in Microsoft Excel to improve data quality. These included:

- Removing duplicate records.
- Checking for blank or missing values.
- Standardizing date formats.
- Verifying data consistency across all columns.
- Creating additional analytical columns to support reporting and visualization.

After cleaning, the dataset was imported into PostgreSQL for exploratory data analysis and later connected to Microsoft Power BI to create an interactive sales performance dashboard.

Figure 1: Overview of the Cleaned Superstore Dataset in Microsoft Excel.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Box
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chair
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Table
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furniture
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Apparel
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Furniture
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Binders
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders
15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Apparel
16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders
17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Krlc	Consumer	United States	Madison	Wisconsin	53721	Central	OFF-ST-10004186	Office Supplies	Storage
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10001007	Office Supplies	Storage
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders
22	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art
23	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Apparel
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chair
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Table
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories

Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Month Name	Weekday	Profit Margin (%)	Sales Category
Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136	November	Tuesday	16	Medium
Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582	November	Tuesday	30	High
Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714	June	Sunday	47	Low
Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031	October	Sunday	-40	High
Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164	October	Sunday	11.25	Low
Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694	June	Monday	29	Low
Art	Neuwell 322	7.28	4	0	1.9656	June	Monday	27	Low
Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152	June	Monday	10	High
Binders	DXL Angle-View Binders with Locking Rings by Samsill	18.504	3	0.2	5.7825	June	Monday	31.25	Low
Appliances	Belkin F5C206VTE1 6 Outlet Surge	114.9	5	0	34.47	June	Monday	30	Low
Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092	June	Monday	5	High
Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568	June	Monday	7.5	High
Paper	Xerox 1867	15.552	3	0.2	5.4432	April	Saturday	35	Low
Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922	December	Monday	32.5	Medium
Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	5	0.8	-123.858	November	Sunday	-180	Low
Binders	Storex DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.816	November	Sunday	-150	Low
Storage	Stur-D-Store Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	6	0	13.3176	November	Tuesday	2	High
Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99	May	Tuesday	18	Low
Art	Neuwell 341	8.56	2	0	2.4824	August	Wednesday	29	Low
Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011	August	Wednesday	7.5	Medium
Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384	August	Wednesday	32.5	Low
Art	Neuwell 318	19.46	7	0	5.0596	December	Friday	26	Low
Appliances	Acco Six-Outlet Power Strip, 4' Cord Length	60.34	7	0	15.6884	December	Friday	26	Low
Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0196	July	Sunday	-1.428571429	Low
Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649	September	Friday	23	High
Binders	Wilson Jones Active Use Binders	11.648	2	0.2	4.2224	January	Saturday	36.25	Low
Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741	January	Saturday	13	Low

3. METHODOLOGY

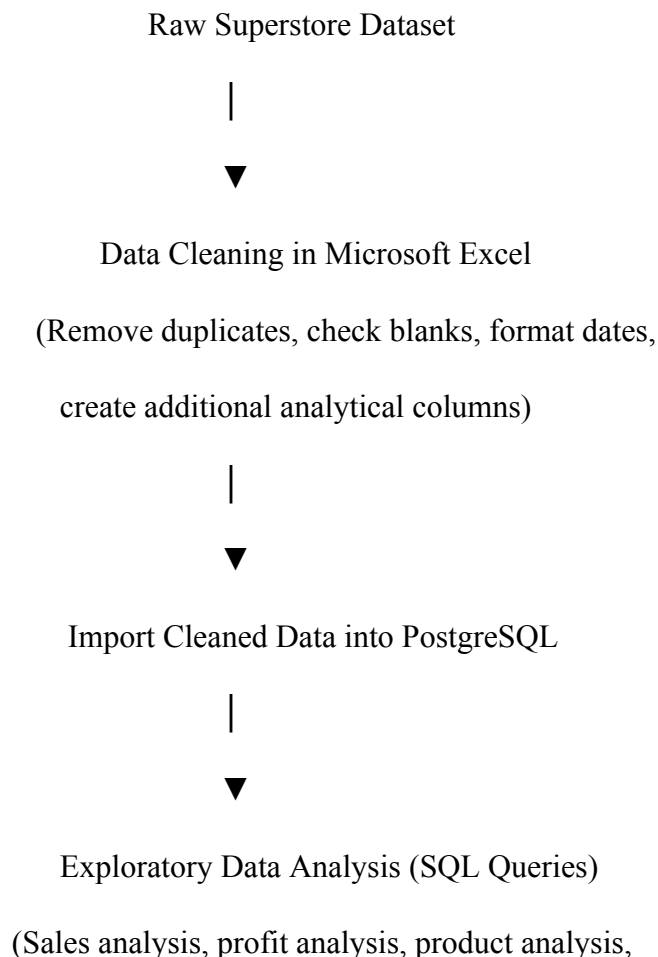
3.1 Overview

This project followed a structured data analytics workflow that transformed raw sales data into meaningful business insights through data cleaning, database management, exploratory data analysis, and dashboard development. Three primary tools were used throughout the project: **Microsoft Excel**, **PostgreSQL**, and **Microsoft Power BI**.

The methodology ensured that the data was cleaned, validated, analyzed, and visualized in a systematic manner, resulting in an interactive dashboard capable of supporting business decision making.

The workflow adopted for this project is illustrated below.

Figure 2: Project Workflow



regional analysis and customer segmentation)

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Connect PostgreSQL Database to Power BI

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Create DAX Measures and Data Model in Power BI

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Develop Interactive Dashboard and Visualizations

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Generate Business Insights & Recommendations

3.2 Data Collection

The project utilized the **Superstore Sales Dataset**, which contains historical retail sales transactions. The dataset includes information relating to customers, products, orders, sales, profit, discounts, shipping details, and geographical locations.

The dataset served as the primary source of information for all analyses performed during the project.

3.3 Data Cleaning Using Microsoft Excel

The first stage of the project involved preparing the dataset in Microsoft Excel to improve data quality before importing it into PostgreSQL.

The following data cleaning tasks were performed:

- Duplicate records were identified and removed to prevent repeated transactions from affecting the analysis.
- Blank or missing values were checked to ensure data completeness.
- Date values were standardized into a consistent format suitable for analysis.
- Additional analytical columns were created where necessary to support reporting.
- The cleaned dataset was reviewed to ensure consistency and accuracy before export.

After the cleaning process was completed, the dataset was saved as a CSV file for import into PostgreSQL.

3.4 Database Management Using PostgreSQL

The cleaned CSV dataset was imported into PostgreSQL to provide a structured database environment for analysis.

Within PostgreSQL, the dataset was successfully imported into the **superstore_cleaned** table. SQL was then used to validate the imported records and perform exploratory data analysis.

Several SQL functions and clauses were applied throughout the analysis, including:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

These queries were used to calculate key business metrics such as total sales, total profit, total orders, average order value, category performance, regional profitability, and product performance.

3.5 Dashboard Development Using Microsoft Power BI

Following SQL analysis, PostgreSQL was connected directly to Microsoft Power BI.

The imported dataset was then used to build an interactive dashboard capable of presenting business performance through multiple visualizations.

Several DAX measures were created, including:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value

These measures were used to develop KPI cards and support the calculations displayed throughout the dashboard.

The dashboard includes:

- KPI Cards
- Monthly Sales Trend
- Sales by Category
- Profit by Region
- Customer Segment Distribution
- Top 10 Products by Sales
- Interactive Slicers (Year, Month, Region, Category, and Segment)

These visualizations enable users to interact with the data and explore sales performance across different business dimensions.

3.6 Business Insight Generation

The final stage of the project involved interpreting the dashboard findings to identify patterns and trends in business performance.

Insights were generated by analyzing:

- Overall sales performance
- Regional profitability
- Product category performance
- Customer segment contribution
- Monthly sales trends
- Top performing products

Based on these findings, practical recommendations were proposed to support strategic decision making and improve business performance.

Figure 3: End to End Analytics Process

Raw Data (Excel)

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Data Cleaning

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CSV Export

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PostgreSQL Database

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SQL Analysis (EDA)

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Power BI Connection

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DAX Measures

|



Interactive Dashboard

|



Business Insights

|



Recommendations

4. DATA CLEANING USING MICROSOFT EXCEL

4.1 Overview

Data cleaning is one of the most important stages in the data analytics process because the quality of the analysis depends on the quality of the data. Raw datasets often contain duplicate records, inconsistent formatting, missing values, and other issues that can lead to inaccurate results if not addressed.

For this project, Microsoft Excel was used as the primary tool for preparing the Superstore Sales Dataset before it was imported into PostgreSQL. The objective was to ensure that the dataset was accurate, complete, and suitable for analysis and visualization.

Several data cleaning techniques were applied to improve the reliability of the dataset.

4.2 Importing the Dataset into Microsoft Excel

The Superstore Sales Dataset was first opened in Microsoft Excel. The dataset contained multiple columns describing customer orders, products, sales, profits, shipping details, and geographical information.

After loading the dataset, each column was reviewed to understand the type of information it contained and to identify any inconsistencies that could affect subsequent analysis.

4.3 Removing Duplicate Records

Duplicate records can distort business metrics such as total sales, profit, and order count by counting the same transaction more than once.

To prevent this issue, the dataset was examined for duplicate entries using Excel's **Remove Duplicates** feature.

The following steps were performed:

1. Selected the entire dataset.

2. Navigated to the **Data** tab.
3. Clicked **Remove Duplicates**.
4. Selected all relevant columns for duplicate detection.
5. Removed duplicate records where applicable.

This process ensured that each transaction in the dataset represented a unique record.

4.4 Checking for Missing Values

Incomplete data can reduce the accuracy of analysis and lead to misleading conclusions.

The dataset was carefully inspected for blank or missing values across all columns.

This involved:

- Reviewing each column for empty cells.
- Confirming that mandatory fields such as **Order ID**, **Order Date**, **Sales**, **Profit**, **Category**, and **Region** contained valid data.
- Verifying that no critical information was missing before proceeding to the analysis stage.

No significant missing values that would affect the analysis were identified after the cleaning process.

4.5 Standardizing Date Formats

The **Order Date** column is essential for analyzing monthly and yearly sales trends. To ensure consistency, the date values were converted into a standardized date format recognized by both PostgreSQL and Power BI.

Standardizing the date format made it possible to:

- Sort records chronologically.
- Create monthly sales trend charts.
- Filter data by year and month.

- Perform time series analysis in Power BI.

4.6 Creating Additional Analytical Columns

To enhance reporting and dashboard development, additional columns were created in Excel.

These included:

- **Month:** Extracted from the Order Date to support monthly trend analysis.
- **Profit Margin:** Used to evaluate the relationship between sales and profit.
- **Sales Category:** Classified sales values into High, Medium, and Low categories for easier comparison.

These derived columns simplified subsequent analysis and visualization in Power BI.

4.7 Verifying Data Quality

Before exporting the dataset, a final review was conducted to ensure that:

- Duplicate records had been removed.
- Blank values had been addressed.
- Date formats were consistent.
- Numeric columns contained valid values.
- Text fields were correctly formatted.
- All records were complete and suitable for import into PostgreSQL.

This validation step helped reduce the likelihood of import errors and ensured the integrity of the data throughout the project.

4.8 Exporting the Cleaned Dataset

Once the cleaning process was completed, the dataset was saved in **CSV (Comma Separated Values)** format.

Saving the file as a CSV allowed it to be imported into PostgreSQL without compatibility issues and provided a structured format for SQL analysis.

The cleaned dataset served as the foundation for the remaining stages of the project, including database management, exploratory data analysis, and dashboard development.

Figure 4: Cleaned Superstore Dataset in Microsoft Excel

superstore_cleaned.xlsx - Excel

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1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
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3	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs
4	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Fort Lauderdale	California	90038	West	OFF-LA-10000577	Office Supplies	Labels
5	4	US-2015-108966	10/11/2015	10/15/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	West	FUR-TA-10000074	Furniture	Tables
6	5	US-2015-108966	10/11/2015	10/15/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	West	OFF-ST-10000760	Office Supplies	Storage
7	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings
8	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Arts
9	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
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11	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances
12	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables
13	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones
14	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper
15	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders
16	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances
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19	18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10001007	Office Supplies	Storage
20	19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10002311	Office Supplies	Arts
21	20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones
22	21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders
23	22	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Arts
24	23	CA-2016-137330														

Figure 4 shows the cleaned version of the Superstore dataset after duplicate records were removed, missing values were checked, date formats were standardized, and additional analytical columns were created. This prepared dataset was subsequently exported as a CSV file and imported into PostgreSQL for further analysis.

5. DATA ANALYSIS USING POSTGRESQL

5.1 Overview

After cleaning and preparing the Superstore Sales Dataset in Microsoft Excel, the next stage of the project involved importing the dataset into **PostgreSQL** for database management and exploratory data analysis (EDA). PostgreSQL is a powerful open source relational database management system (RDBMS) that enables efficient storage, querying, and analysis of large datasets using Structured Query Language (SQL).

Using PostgreSQL allowed the dataset to be organized in a structured database environment, making it easier to retrieve information, calculate business metrics, and identify meaningful trends that would later be visualized in Microsoft Power BI.

5.2 Creating the Database

The first step was to create a new PostgreSQL database to store the cleaned Superstore dataset.

A database named **Superstore** was created in pgAdmin, providing a dedicated workspace for managing the project data. Within this database, a table named **superstore_cleaned** was created to store the imported sales records.

Creating a separate database helped organize the project and ensured that all SQL operations were performed within a structured environment.

5.3 Importing the Dataset

After completing the data cleaning process in Microsoft Excel, the dataset was saved in **CSV (Comma Separated Values)** format to ensure compatibility with PostgreSQL.

The CSV file was imported into the **superstore_cleaned** table using pgAdmin's **Import/Export Data** feature. During the import process, the appropriate file path, delimiter (comma), and header options were specified to ensure that the data was imported correctly.

Several import challenges were encountered during this stage, including file path errors and CSV formatting issues. These problems were resolved by correcting the file path, saving the dataset in the proper CSV format, and verifying the import settings. Once these issues were addressed, the dataset was successfully imported into PostgreSQL.

5.4 Data Validation

Following the import process, the dataset was validated to ensure that all records had been imported successfully and accurately.

Basic SQL queries were executed to verify the contents of the table and confirm that the data matched the cleaned Excel dataset.

Examples of validation queries included:

- Displaying all records in the table.
- Counting the total number of records.
- Checking that important columns contained valid values.
- Reviewing sample records for consistency.

This validation step confirmed that the database was ready for further analysis.

5.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed using SQL queries to summarize the dataset and answer key business questions.

The analysis focused on measuring sales performance, profitability, customer purchasing behavior, regional performance, and product performance.

Several SQL functions and clauses were used throughout the analysis, including:

- **SELECT** – Retrieved specific columns and records from the dataset.
- **WHERE** – Filtered records based on specified conditions.

- **GROUP BY** – Grouped records for summarized analysis.
- **ORDER BY** – Sorted query results in ascending or descending order.
- **SUM()** – Calculated total sales and total profit.
- **COUNT()** – Counted the total number of orders and records.
- **AVG()** – Calculated average sales and average order value.

These SQL operations enabled efficient analysis of the retail sales data and produced the metrics required for dashboard development.

5.6 Business Questions Answered Using SQL

SQL queries were used to answer several important business questions, including:

- What is the total sales revenue generated by the business?
- What is the total profit earned?
- How many customer orders were processed?
- What is the average order value?
- Which product categories generate the highest sales?
- Which regions generate the highest profit?
- Which customer segment contributes the most revenue?
- Which products are the top selling products?
- How does sales performance change over time?

The results obtained from these queries formed the basis for the key performance indicators (KPIs) and visualizations developed in Microsoft Power BI.

5.7 SQL Queries Used in the Analysis

Throughout the project, SQL queries were written to retrieve and summarize data from the **superstore_cleaned** table.

Typical queries included:

- Retrieving all records from the table.
- Counting the total number of orders.
- Calculating total sales and profit.
- Calculating average sales values.
- Grouping sales by category and region.
- Ranking products based on sales performance.
- Summarizing customer segment performance.

These queries transformed raw transactional data into meaningful business information that supported informed decision making.

5.8 Contribution of PostgreSQL to the Project

Using PostgreSQL provided several advantages during the project:

- Improved organization and storage of the cleaned dataset.
- Faster querying of large amounts of data.
- Accurate calculation of business metrics using SQL.
- Easy integration with Microsoft Power BI.
- Reliable validation of data before visualization.

The use of PostgreSQL demonstrated how relational databases can support business intelligence by providing a structured environment for data analysis and reporting.

Figure 5: Superstore Dataset Imported into PostgreSQL

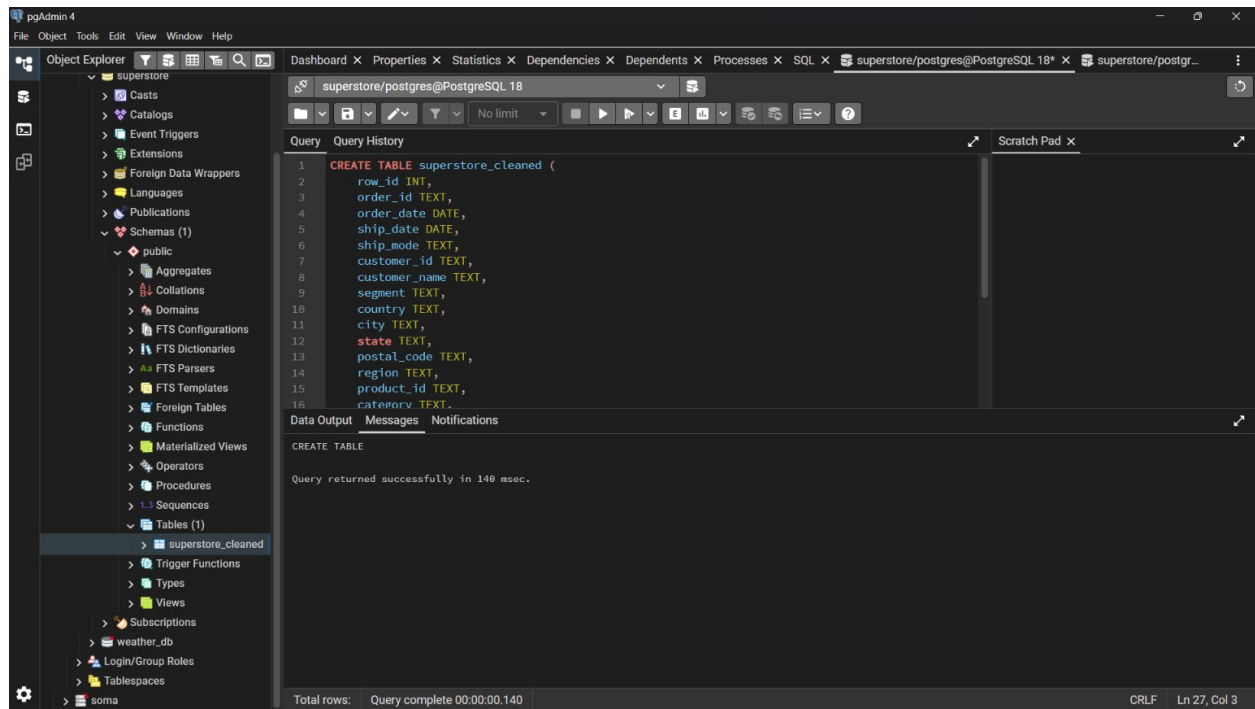


Figure 5. The Superstore Sales dataset successfully imported into PostgreSQL (pgAdmin), where SQL queries were used to perform exploratory data analysis and generate key business metrics.

6. DASHBOARD DEVELOPMENT USING POWER BI

After cleaning and preparing the dataset in Microsoft Excel and storing it in PostgreSQL, the final stage of the project involved creating an interactive dashboard in Microsoft Power BI. The dashboard was designed to present key business metrics visually, making it easier to monitor sales performance, profitability, customer segments, regional performance, and product trends.

6.1 Connecting PostgreSQL to Power BI

The cleaned dataset stored in PostgreSQL was connected directly to Microsoft Power BI.

The connection process involved:

- Opening Power BI Desktop.
- Selecting **Get Data** → **PostgreSQL Database**.
- Entering the PostgreSQL server name and database name.
- Authenticating using the PostgreSQL username and password.
- Importing the **public.superstore_cleaned** table into Power BI.

Connecting directly to PostgreSQL ensured that Power BI could access the cleaned and structured data for visualization and analysis.

6.2 Creating DAX Measures

To summarize business performance, several DAX (Data Analysis Expressions) measures were created.

Total Sales

This measure calculates the total revenue generated from all orders.

Total Sales =

SUM('public superstore_cleaned'[sales])

Purpose:

Displays the overall sales generated by the company.

Total Profit

This measure calculates the total profit earned from all sales.

Total Profit =

`SUM('public superstore_cleaned'[profit])`

Purpose:

Shows the company's profitability after sales.

Total Orders

This measure counts the number of unique customer orders.

Total Orders =

`DISTINCTCOUNT('public superstore_cleaned'[order_id])`

Purpose:

Determines how many orders were processed.

Average Order Value

This measure calculates the average revenue generated per order.

Average Order Value =

`DIVIDE([Total Sales], [Total Orders],0)`

Purpose:

Measures the average amount customers spend on each order.

6.3 Building Visualizations

Several visualizations were developed to communicate business insights effectively.

KPI Cards

Four KPI cards were created at the top of the dashboard to display the most important business metrics.

The cards display:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value

Purpose

These KPIs provide users with a quick overview of overall business performance without needing to analyze multiple charts.

Monthly Sales Trend (Line Chart)

A line chart was created to display **Total Sales by Month**.

Purpose

This visualization helps identify sales trends throughout the year, making it easy to recognize seasonal patterns, sales growth, and months with higher or lower performance.

Total Sales by Category (Bar Chart)

A horizontal bar chart was created to compare sales across product categories.

The categories include:

- Technology

- Furniture
- Office Supplies

Purpose

This visualization highlights which product category contributes the highest revenue and allows easy comparison among categories.

Total Profit by Region (Column Chart)

A column chart was created to compare profits across different sales regions.

The regions include:

- West
- East
- South
- Central

Purpose

This chart identifies the most profitable and least profitable regions, helping businesses evaluate regional performance.

Total Sales by Product (Bar Chart)

A horizontal bar chart was created to display the best performing products based on sales.

Purpose

This visualization identifies the products generating the highest revenue and helps management understand which products contribute most to overall sales.

Sales by Customer Segment (Donut Chart)

A donut chart was created to show the proportion of sales contributed by each customer segment.

The customer segments include:

- Consumer
- Corporate
- Home Office

Purpose

This chart provides an overview of how sales are distributed among different customer groups and identifies the largest revenue generating segment.

Interactive Slicers

Several slicers were added to improve dashboard interactivity.

The dashboard includes filters for:

- Year
- Month
- Region
- Category
- Customer Segment

Purpose

These slicers allow users to dynamically filter the dashboard and analyze sales performance from different perspectives without modifying the underlying data.

7. DASHBOARD OVERVIEW

This section presents the Power BI dashboards developed for the Superstore Sales Analysis project. The dashboards provide an interactive overview of sales performance, profitability, customer segmentation, product performance, and business insights.

Figure 3. Executive Sales Dashboard

Description

The Executive Sales Dashboard provides a high level overview of the company's sales performance using Key Performance Indicators (KPIs) and interactive visualizations.

Dashboard Components

- KPI Cards
 - Total Sales: **\$2.30M**
 - Total Profit: **\$286.40K**
 - Total Orders: **5,009**
 - Average Order Value: **\$458.61**
- Monthly Sales Trend
- Sales by Category
- Profit by Region
- Top Selling Products

Purpose

This dashboard enables decision makers to quickly monitor business performance, identify profitable categories and regions, and understand monthly sales trends.

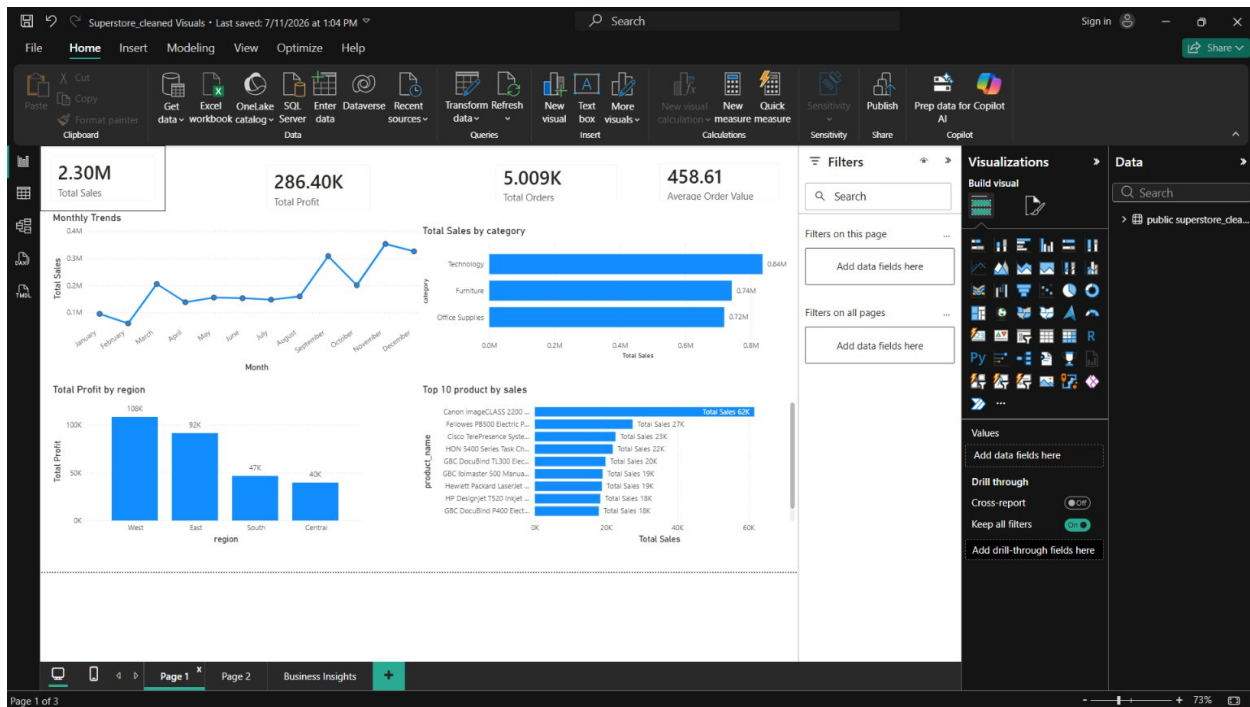


Figure 4. Interactive Filter Dashboard

Description

The Interactive Dashboard enables users to dynamically explore the sales dataset using slicers.

Available Filters

- Year
- Month
- Region
- Category
- Customer Segment

The donut chart automatically updates based on the selected filters, allowing users to analyze customer segment contributions under different conditions.

Purpose

This dashboard improves user interaction and allows detailed exploration of the sales data from multiple perspectives.

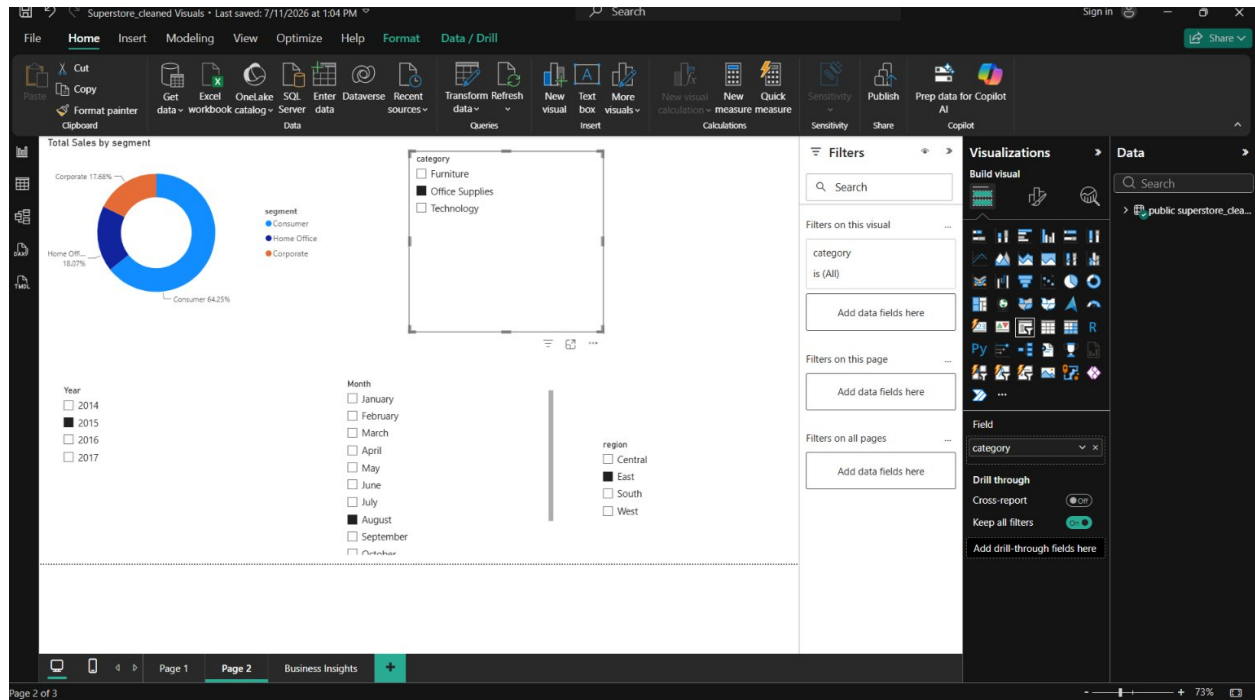


Figure 5. Business Insights Dashboard

Description

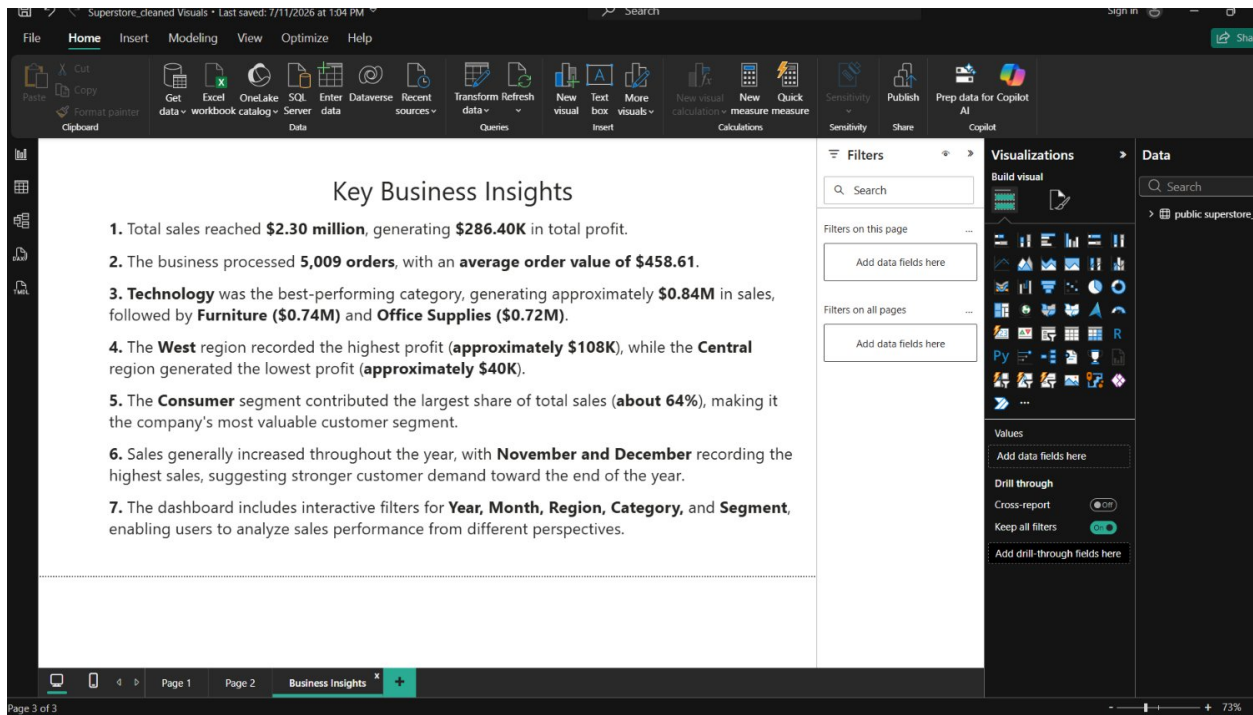
The Business Insights page summarizes the major findings from the sales analysis.

The dashboard highlights:

- Total Sales achieved
- Total Profit generated
- Total Orders processed
- Average Order Value
- Highest performing Category
- Most profitable Region
- Largest Customer Segment
- Monthly Sales Trend

- Interactive filtering capabilities

These insights provide management with concise and actionable information to support business decisions.



8. KEY FINDINGS

The analysis of the Superstore Sales dataset provided valuable insights into the company's overall sales performance, profitability, customer purchasing behavior, and product performance. The key performance indicators (KPIs) and dashboard visualizations highlight the most important findings derived from the analysis.

8.1 Key Performance Indicators (KPIs)

The table below summarizes the primary business metrics calculated during the project.

KPI	Result	Description
Total Sales	\$2.30 Million	Total revenue generated from all sales transactions.
Total Profit	\$286.40 Thousand	Total profit earned after accounting for costs and discounts.
Total Orders	5,009	Total number of unique customer orders processed.
Average Value	Order \$458.61	Average revenue generated per customer order.

Table 1: *Summary of the key performance indicators calculated from the Superstore Sales dataset.*

8.2 Sales Performance

The analysis showed that the Superstore generated approximately \$2.30 million in total sales during the reporting period. This indicates a strong sales volume and demonstrates the company's ability to generate consistent revenue across different products, customer segments, and regions.

The monthly sales trend further revealed fluctuations in sales throughout the year, with higher sales recorded during the final months of the year, suggesting the influence of seasonal demand.

8.3 Profitability

Although the company generated substantial revenue, the total profit amounted to approximately \$286.40 thousand, which is significantly lower than total sales. This suggests that factors such as discounts, operating costs, or product pricing may have affected overall profitability.

The profit analysis also highlighted differences in performance across regions, indicating opportunities for improving profitability in lower performing areas.

8.4 Product Category Performance

Among the three product categories analyzed:

- Technology generated the highest sales.
- Office Supplies contributed steady sales across the reporting period.
- Furniture generated comparatively lower sales and may require further evaluation to improve performance.

This finding suggests that technology products are the primary revenue drivers for the business.

8.5 Regional Performance

The regional analysis indicated differences in profitability across geographical locations.

The West region generated the highest profit, making it the strongest performing region during the analysis period. This suggests effective sales performance and stronger customer demand within that market.

Other regions contributed positively to overall sales but demonstrated comparatively lower profitability.

8.6 Customer Segment Analysis

The dashboard showed that the Consumer segment contributed the largest share of total sales.

This indicates that individual consumers represent the company's largest customer base and should remain a key focus for marketing strategies and customer retention initiatives.

The Corporate and Home Office segments also contributed to sales but accounted for smaller proportions of total revenue.

8.7 Product Performance

The product analysis identified several high performing products that generated substantial revenue for the business.

Understanding which products contribute the most to sales enables management to:

- Prioritize inventory management.
- Improve demand forecasting.
- Develop targeted marketing campaigns.
- Optimize product offerings.

8.8 Overall Findings

The findings from this analysis demonstrate that the Superstore business performs strongly in terms of revenue generation while presenting opportunities to improve profitability and optimize performance across certain regions and product categories.

The integration of Microsoft Excel, PostgreSQL, SQL, and Power BI made it possible to transform raw transactional data into meaningful business insights that can support informed decision making.

Figure 6: Executive Dashboard

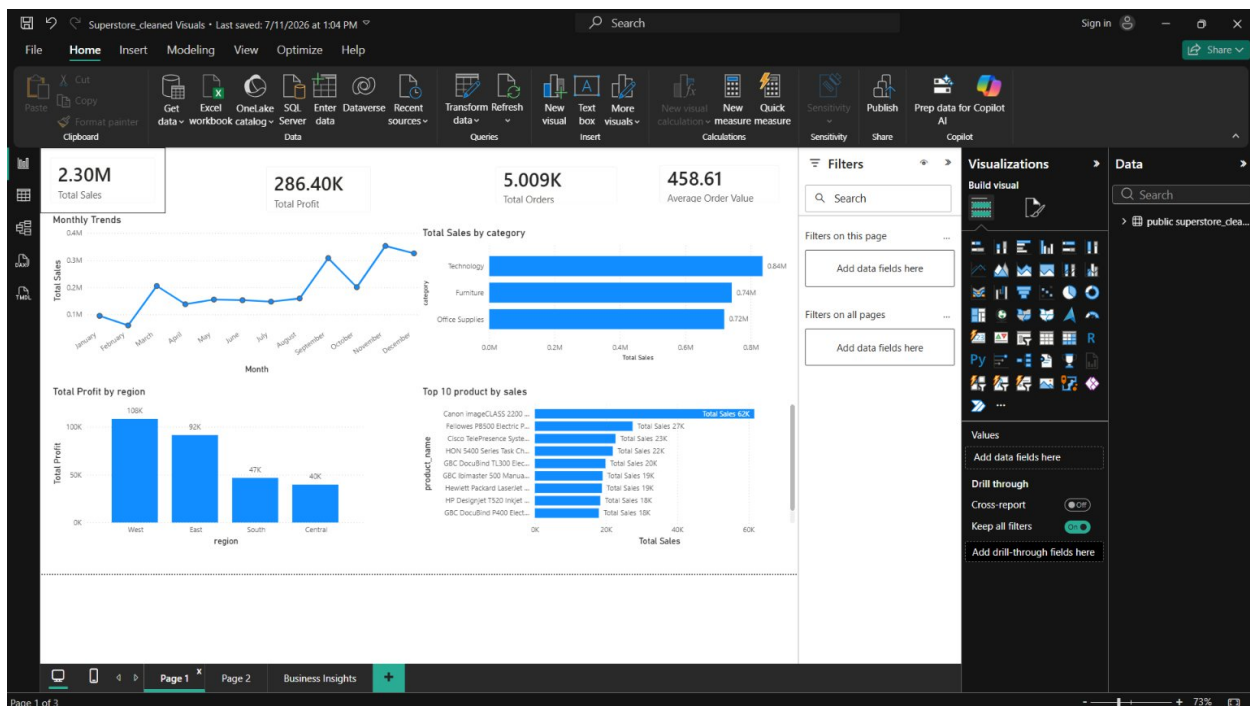


Figure 6. *Executive dashboard displaying the key performance indicators and visualizations used to evaluate sales performance, profitability, customer segments, product categories, and regional performance.*

9. BUSINESS INSIGHTS

The interactive Power BI dashboard provided valuable insights into the company's sales performance, profitability, customer behavior, and regional performance. By analyzing the key performance indicators and visualizations, several important business insights were identified.

9.1 Overall Business Performance

The analysis revealed that the business generated approximately **\$2.30 million** in total sales from **5,009 customer orders**, resulting in a total profit of approximately **\$286.40 thousand**. While the company achieved strong sales revenue, the difference between total sales and total profit suggests that profitability may be influenced by factors such as discounts, product costs, or pricing strategies.

Overall, the business demonstrates healthy sales performance with opportunities to further improve profit margins.

9.2 Technology Products Drive Revenue Growth

The dashboard shows that the **Technology** category generated the highest sales among all product categories.

This indicates that customers have a strong demand for technology related products, making this category the company's primary source of revenue.

This insight suggests that management should continue investing in technology products by ensuring adequate inventory levels, expanding product offerings, and implementing targeted marketing campaigns.

9.3 The West Region Is the Most Profitable Market

Regional analysis revealed that the **West** region generated the highest profit compared to the East, South, and Central regions.

This suggests that business operations, customer demand, and product performance are strongest within this region.

Understanding the factors contributing to the West region's success may help the company replicate similar strategies in other regions with lower profitability.

9.4 Consumer Customers Contribute the Largest Share of Sales

The customer segment analysis indicates that the **Consumer** segment contributes the highest proportion of total sales.

This finding suggests that individual consumers represent the company's largest customer base and play a significant role in overall business performance.

Maintaining strong relationships with consumer customers through loyalty programs, personalized promotions, and quality customer service may help sustain future sales growth.

9.5 Sales Increase During the End of the Year

The Monthly Sales Trend visualization shows noticeable growth in sales during the final months of the year, particularly **November and December**.

This seasonal increase may be associated with holiday shopping, promotional campaigns, or year end purchasing behavior.

Recognizing this trend allows management to prepare for periods of increased demand by improving inventory planning, staffing, and marketing activities.

9.6 Interactive Dashboard Supports Better Decision Making

The dashboard includes interactive slicers that allow users to filter the data by:

- Year
- Month

- Region
- Category
- Customer Segment

These filters enable users to explore business performance from different perspectives without modifying the underlying dataset.

The interactive nature of the dashboard makes it an effective business intelligence tool for monitoring performance and supporting strategic decision making.

9.7 Business Value of the Dashboard

The completed dashboard transforms raw sales data into meaningful visual information that can be easily understood by managers and stakeholders.

Instead of manually reviewing thousands of transaction records, users can quickly identify:

- Overall business performance
- High performing product categories
- Regional profitability
- Customer purchasing patterns
- Monthly sales trends
- Key performance indicators

This improves reporting efficiency and enables faster, evidence based business decisions.

Figure 7: Business Insights Dashboard

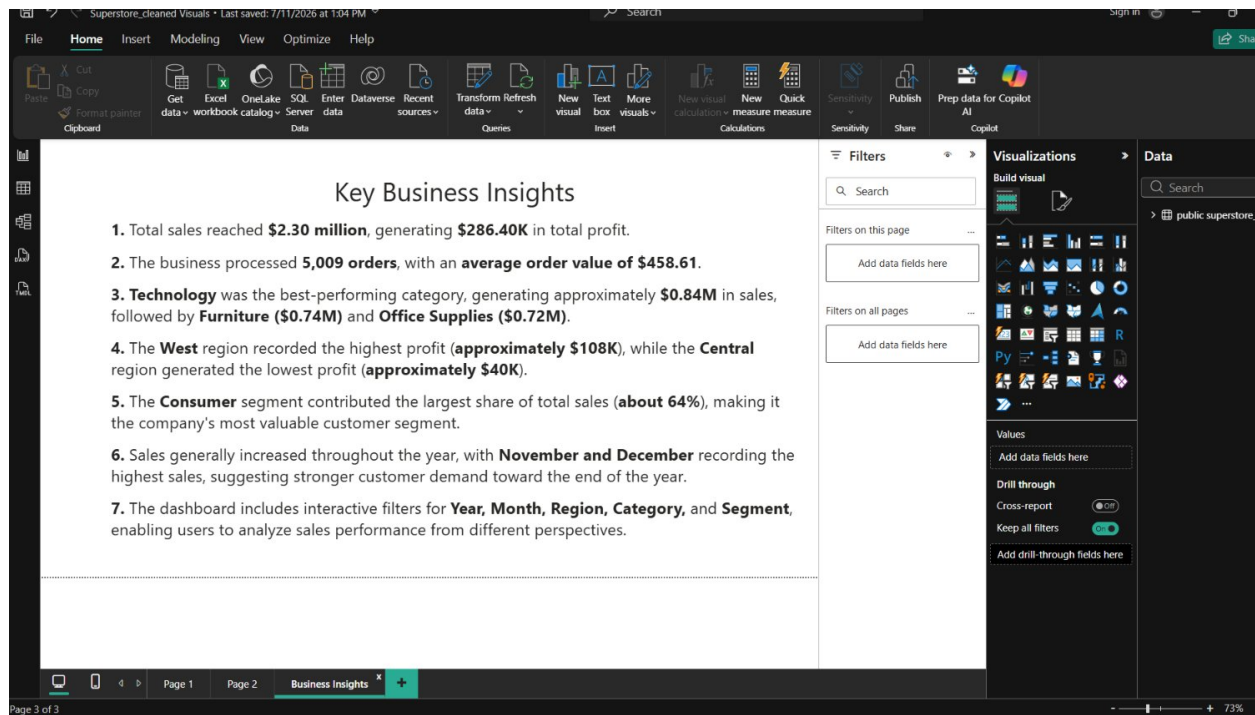


Figure 7. *Business Insights dashboard summarizing the key findings from the Superstore Sales Analysis, including overall performance, category analysis, regional profitability, customer segmentation, and seasonal sales trends.*

10. RECOMMENDATIONS

Based on the findings obtained from the analysis of the Superstore Sales dataset, several recommendations can be made to improve business performance, increase profitability, and support strategic decision making.

10.1 Increase Investment in High Performing Product Categories

The analysis showed that the **Technology** category generated the highest sales among all product categories. This indicates strong customer demand and highlights Technology as a key driver of revenue.

The company should consider expanding its range of technology products, maintaining adequate inventory levels, and implementing targeted marketing campaigns to capitalize on customer demand. Investing in high performing product categories can help sustain revenue growth and improve overall business performance.

10.2 Improve Profitability in Lower Performing Regions

Although all regions contributed to overall sales, the analysis revealed differences in profitability across regions. While the **West** region recorded the highest profit, other regions showed comparatively lower performance.

Management should investigate the factors affecting profitability in these regions. Areas for review may include pricing strategies, operational costs, product availability, customer preferences, and promotional activities. Addressing these factors could improve profitability and create a more balanced performance across all regions.

10.3 Strengthen Customer Retention Strategies

The dashboard indicates that the **Consumer** segment contributed the largest share of total sales, making it the company's most valuable customer group.

To maintain and grow this customer base, the company should implement customer retention initiatives such as loyalty programmes, personalized promotions, and excellent customer service.

Retaining existing customers is often more cost effective than acquiring new ones and can contribute to long term business growth.

10.4 Prepare for Seasonal Demand

The monthly sales trend showed increased sales activity towards the end of the year, particularly during **November and December**. This pattern suggests seasonal demand that may be associated with holiday shopping and promotional events.

The company should prepare for these peak periods by increasing inventory, ensuring adequate staffing, and planning marketing campaigns in advance. Effective preparation can help prevent stock shortages, improve customer satisfaction, and maximize sales opportunities during high demand periods.

10.5 Use Business Intelligence for Continuous Monitoring

The interactive Power BI dashboard developed in this project provides a practical tool for monitoring business performance in real time. Rather than relying on static reports, management can use the dashboard to track key performance indicators, evaluate sales trends, and identify changes in customer behaviour.

Regular use of the dashboard will support faster, evidence based decision making and enable the business to respond proactively to emerging trends and performance issues.

10.6 Continue Data Quality Management

The success of business analysis depends on the quality of the underlying data. Maintaining accurate, complete, and consistent data should remain a priority.

The company should establish routine data quality checks to identify duplicate records, missing values, and formatting inconsistencies before data is used for reporting or analysis. Consistent data management practices will improve the reliability of future analyses and dashboards.

10.7 Expand Future Analysis

This project focused primarily on historical sales performance. Future analyses could be extended to provide deeper business insights by incorporating additional data sources and advanced analytical techniques.

Potential areas for future work include:

- Sales forecasting using historical trends.
- Customer lifetime value analysis.
- Product demand forecasting.
- Profit margin analysis by product.
- Customer churn analysis.
- Predictive analytics using machine learning models.

These advanced analyses could provide management with more proactive decision making tools and support long term business planning.

11. CHALLENGES ENCOUNTERED

Throughout the development of this project, several technical challenges were encountered during the data cleaning, database management, and dashboard development stages. Addressing these challenges provided valuable learning opportunities and strengthened my problem solving skills.

11.1 Data Cleaning Challenges in Microsoft Excel

One of the initial challenges was ensuring that the dataset was suitable for analysis. The dataset was reviewed to identify duplicate records and blank values that could affect the accuracy of the analysis.

Excel's **Remove Duplicates** feature was used to eliminate duplicate records, while the dataset was carefully examined for missing values and formatting inconsistencies. Date fields were also standardized to ensure compatibility with PostgreSQL and Microsoft Power BI.

These data preparation steps improved the quality and reliability of the dataset before it was imported into the database.

11.2 Challenges Importing Data into PostgreSQL

Importing the cleaned dataset into PostgreSQL proved to be one of the most challenging stages of the project.

Several issues were encountered during the import process, including:

- Incorrect CSV file paths.
- Saving the dataset in the wrong file format.
- Errors caused by incorrect file extensions.
- Import process failures due to configuration settings.

These challenges were resolved by:

- Saving the cleaned dataset as a valid **CSV (Comma Separated Values)** file.
- Verifying the correct file path before importing.

- Configuring the import settings correctly in pgAdmin.
- Repeating the import process until the dataset was successfully loaded into the **public.superstore_cleaned** table.

Successfully resolving these issues improved my understanding of data import procedures in PostgreSQL.

11.3 Database and SQL Challenges

While working in PostgreSQL, several SQL related issues were encountered.

One common error occurred when attempting to create a table that already existed, resulting in the following message:

ERROR: relation "superstore_cleaned" already exists

This issue occurred because the table had already been created. The problem was resolved by querying the existing table instead of attempting to recreate it.

Additional challenges included:

- Verifying that data had been imported successfully.
- Understanding PostgreSQL schemas.
- Referencing the correct table (public.superstore_cleaned) in SQL queries.
- Validating imported records before proceeding to dashboard development.

These experiences improved my understanding of relational databases and SQL query execution.

11.4 Connecting PostgreSQL to Microsoft Power BI

Establishing a connection between PostgreSQL and Microsoft Power BI presented another challenge.

Initially, the required table was not visible in Power BI because of connection and navigation issues. After reviewing the database configuration, refreshing the connection, and selecting the correct schema and table, the dataset was successfully imported into Power BI.

This process demonstrated the importance of verifying database connections and ensuring that the correct data source is selected.

11.5 Creating DAX Measures

Developing DAX measures required careful attention to table names, column names, and syntax.

During this stage, errors such as **"Cannot find table 'superstore_cleaned'"** were encountered because the DAX formulas referenced an incorrect table name.

The issue was resolved by using the correct table name imported into Power BI (**public superstore_cleaned**) and verifying that each column reference matched the imported dataset.

Successfully creating measures for **Total Sales**, **Total Profit**, **Total Orders**, and **Average Order Value** improved my understanding of DAX and Power BI calculations.

11.6 Dashboard Design and Visualization

Designing an effective dashboard required selecting appropriate visualizations and arranging them in a way that clearly communicated business performance.

Challenges encountered during this stage included:

- Choosing suitable chart types.
- Formatting KPI cards.
- Sorting charts correctly.
- Configuring slicers.
- Customizing chart titles.
- Organizing visuals into a clear and professional layout.

These challenges were addressed through iterative design improvements, resulting in a dashboard that is visually appealing, interactive, and easy to interpret.

11.7 Lessons Learned

Overcoming these challenges significantly improved my technical skills and confidence in using Microsoft Excel, PostgreSQL, SQL, and Microsoft Power BI.

This project provided practical experience in:

- Data cleaning and preparation.
- Database creation and management.
- SQL querying and troubleshooting.
- Connecting databases to Power BI.
- Writing DAX measures.
- Building interactive dashboards.
- Interpreting data to generate business insights.

Each challenge strengthened my ability to troubleshoot technical issues and reinforced the importance of patience, attention to detail, and systematic problem solving throughout the data analytics process.

12. CONCLUSION

This project successfully demonstrated the complete data analytics process, from data preparation to business intelligence reporting, using Microsoft Excel, PostgreSQL, SQL, and Microsoft Power BI. The Superstore Sales dataset was transformed from raw transactional data into an interactive dashboard that provides meaningful insights into business performance.

The project began with data cleaning in Microsoft Excel, where the dataset was reviewed to remove duplicate records, check for missing values, and standardize data formats. These preprocessing activities ensured that the data was accurate, consistent, and suitable for further analysis.

Following the cleaning stage, the dataset was imported into PostgreSQL, where it was stored in a relational database and analyzed using SQL. Various SQL queries were executed to validate the imported data and calculate key business metrics, including total sales, total profit, total orders, and average order value. Additional analyses were performed to evaluate product categories, customer segments, regional performance, and sales trends.

The cleaned and validated dataset was then connected to Microsoft Power BI for dashboard development. Using DAX measures and interactive visualizations, a business intelligence dashboard was created to present the analysis in a clear and user friendly format. The dashboard allows users to monitor key performance indicators, explore sales trends, compare regional performance, and analyze customer purchasing behavior through interactive filters and charts.

The analysis revealed several important findings. The Technology category generated the highest sales, the West region recorded the highest profit, and the Consumer segment contributed the largest share of revenue. Seasonal trends also indicated stronger sales performance during the final months of the year. These insights demonstrate how data visualization can support strategic planning and informed business decision making.

Beyond the analytical findings, this project strengthened practical skills in data cleaning, relational database management, SQL querying, DAX calculations, dashboard design, and business intelligence reporting. It also highlighted the importance of data quality, structured analysis, and effective visualization in transforming raw data into actionable insights.

Overall, this project demonstrates the practical application of modern data analytics tools to solve business problems and communicate results effectively. It serves as evidence of my ability to complete an end to end analytics project and provides a strong foundation for future work in data analysis, business intelligence, and data driven decision making.

APPENDIX

The appendices provide supplementary materials that support the processes, analyses, and findings presented in this report. They include screenshots, SQL queries, DAX measures, and dashboard visuals that demonstrate the practical implementation of the project.

Appendix A: Data Preparation in Microsoft Excel

This appendix presents the data cleaning and preparation activities carried out in Microsoft Excel before importing the dataset into PostgreSQL. These screenshots demonstrate how the dataset was prepared to ensure accuracy, consistency, and reliability for analysis.

Figure A1. Original Superstore Dataset

File Home Insert Draw Page Layout Formulas Data Review View Developer Help																				Share					
Clipboard		Font					Alignment			Number		Conditional Formatting		Table Styles		Cell Styles		Cells		Formulas		Editing		Add-ins	
Paste		Calibri 11 A A					Wrap Text			General		%		Format as Table		Insert Delete Format		Σ AutoSum Fill Clear		Sort & Filter Find & Select		Add-ins			
Copy		B I U					Merge & Center			\$ %															
Format Painter																									
A1		Row ID																							
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit						
1	CA-2016-1	#####	#####	Second Cl	CG-12520	Claire Gut	Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcases	Bush Some	261.96	2	0	41.9136					
2	CA-2016-1	#####	#####	Second Cl	CG-12520	Claire Gut	Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs	Hon Delux	731.94	3	0	219.582					
3	CA-2016-1	#####	#####	Second Cl	DV-13045	Darrin Van	Corporate	United Sta	Los Angeles	California	90036	West	OFF-LA-10	Office Sup	Labels	Self-Adhes	14.62	2	0	6.8714					
4	US-2015-1	#####	#####	Standard	CO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	FUR-TA-1C	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.031					
5	US-2015-1	#####	#####	Standard	CO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	OFF-ST-10	Office Sup	Storage	Eldon Fold	22.368	2	0.2	2.5164					
6	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	FUR-FU-1C	Furniture	Furnishing	Eldon Expr	48.86	7	0	14.1694					
7	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	OFF-AR-1C	Office Sup	Art	Newell 32	7.28	4	0	1.9656					
8	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1C	Technolog	Phones	Mitel 532C	907.152	6	0.2	90.7152					
9	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	OFF-BI-10	Office Sup	Binders	DXL Angle	18.504	3	0.2	5.7825					
10	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	OFF-AP-10	Office Sup	Appliances	Belkin PSC	114.9	5	0	34.47					
11	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	FUR-TA-1C	Furniture	Tables	Chromcraft	1706.184	9	0.2	85.3092					
12	CA-2014-1	6/9/2014	#####	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1C	Technolog	Phones	Konitel 25	911.424	4	0.2	68.3568					
13	CA-2017-1	#####	#####	Standard	AA-10480	Andrew All	Consumer	United Sta	Concord	North Car	28027	South	OFF-PA-10	Office Sup	Paper	Xerox 196	15.552	3	0.2	5.4432					
14	CA-2016-1	#####	#####	Standard	CM-15070	Irene Mad	Consumer	United Sta	Seattle	Washingtc	98103	West	OFF-BI-10	Office Sup	Binders	Fellowes P	407.976	3	0.2	132.5922					
15	US-2015-1	#####	#####	Standard	CHP-14815	Harold Pav	Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-AP-10	Office Sup	Appliances	Holmes Re	68.81	5	0.8	-123.858					
16	US-2015-1	#####	#####	Standard	CHP-14815	Harold Pav	Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Storex Dur	2.544	3	0.8	-3.816					
17	CA-2014-1	#####	#####	Standard	CPK-19075	Pete Kriz	Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Sup	Storage	Stur-D-Sto	665.88	6	0	13.3176					
18	CA-2014-1	#####	#####	Second Cl	AG-10270	Alejandro	Consumer	United Sta	West Jor	Utah	84084	West	OFF-ST-10	Office Sup	Storage	Fellowes S	55.5	2	0	9.99					
19	CA-2014-1	9/1/2014	Second Cl	ZD-21925	Zuschuss C	Consumer	United Sta	San Franci	California	94109	West	OFF-AR-1C	Office Sup	Art	Newell 34	8.56	2	0	2.4824						
20	CA-2014-1	9/1/2014	Second Cl	ZD-21925	Zuschuss C	Consumer	United Sta	San Franci	California	94109	West	TEC-PH-1C	Technolog	Phones	Cisco SPA	213.48	3	0.2	16.011						
21	CA-2014-1	9/1/2014	Second Cl	ZD-21925	Zuschuss C	Consumer	United Sta	San Franci	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jor	22.72	4	0.2	7.384						
22	CA-2016-1	#####	#####	Standard	KB-16585	Ken Black	Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AR-1C	Office Sup	Art	Newell 311	19.46	7	0	5.0596					
23	CA-2016-1	#####	#####	Standard	KB-16585	Ken Black	Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AP-10	Office Sup	Appliances	Acco Six-C	60.34	7	0	15.6884					
24	US-2017-1	#####	#####	Standard	CF-20605	Sandra Fla	Consumer	United Sta	Philadelph	Pennsylv	19140	East	FUR-CH-1C	Furniture	Chairs	Global Del	71.372	2	0.3	-1.0196					
25	CA-2015-1	#####	#####	Standard	EB-13870	Emily Burn	Consumer	United Sta	Orem	Utah	84057	West	FUR-TA-1C	Furniture	Tables	Bretford C	1044.63	3	0	240.2649					
26	CA-2016-1	#####	#####	Second Cl	EH-13945	Eric Hoffm	Consumer	United Sta	Los Angeles	California	90049	West	OFF-BI-10	Office Sup	Binders	Wilson Jor	11.648	2	0.2	4.2224					
27	CA-2016-1	#####	#####	Second Cl	EH-13945	Eric Hoffm	Consumer	United Sta	Los Angeles	California	90049	West	TEC-AC-1C	Technolog	Accessorie	Imation 8C	90.57	3	0	11.7741					

Ready Scroll Lock Accessibility: Unavailable

Figure A2. Cleaned Dataset

The image displays two screenshots of a Microsoft Excel spreadsheet, both titled 'superstore_cleaned.xlsx'. The top screenshot shows a detailed view of the first 28 rows of the 'Orders' table. The columns are: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, and Sub-Category. The data includes various orders from different customers, such as Claire Gule, Sean O'Donnell, and Brosina Hoffman, across different states like Kentucky, California, and Florida. The bottom screenshot shows a detailed view of the first 28 rows of the 'Products' table. The columns are: Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Month Name, Weekday, Profit Margin (%), and Sales Category. The data includes products like Bookcases, Chairs, Labels, Tables, Storage, Furnishings, Art, Phones, Binders, Appliances, and Accessories, with their respective sales, quantities, and profit margins.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders
15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances
16	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders
17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-18075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10000486	Office Supplies	Storage
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders
22	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art
23	CA-2016-137330	12/9/2016	12/13/2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Appliances
24	US-2017-156099	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories

Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Month Name	Weekday	Profit Margin (%)	Sales Category
Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136	November	Tuesday	16	Medium
Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582	November	Tuesday	30	High
Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714	June	Sunday	47	Low
Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031	October	Sunday	-40	High
Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164	October	Sunday	11.25	Low
Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694	June	Monday	29	Low
Art	Newell 322	7.28	4	0	1.9656	June	Monday	27	Low
Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152	June	Monday	10	High
Binders	DXL Angle-View Binders with Locking Rings by Samsill	18.504	3	0.2	5.7825	June	Monday	31.25	Low
Appliances	Belkin F5C205VTE1 6 Outlet Surge	114.9	5	0	34.47	June	Monday	30	Low
Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092	June	Monday	5	High
Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568	June	Monday	7.5	High
Paper	Xerox 1967	15.552	3	0.2	5.4432	April	Saturday	35	Low
Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922	December	Monday	32.5	Medium
Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	5	0.8	-123.858	November	Sunday	-180	Low
Binders	Storrex DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.816	November	Sunday	-150	Low
Storage	Stur-D-Store Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	6	0	13.3176	November	Tuesday	2	High
Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99	May	Tuesday	18	Low
Art	Newell 341	8.56	2	0	2.4824	August	Wednesday	29	Low
Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011	August	Wednesday	7.5	Medium
Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384	August	Wednesday	32.5	Low
Art	Newell 318	19.46	7	0	5.0596	December	Friday	26	Low
Appliances	Acco Six-Outlet Power Strip, 4' Cord Length	60.34	7	0	15.6884	December	Friday	26	Low
Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0196	July	Sunday	-1.428571429	Low
Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649	September	Friday	23	High
Binders	Wilson Jones Active Use Binders	11.648	2	0.2	4.2224	January	Saturday	36.25	Low
Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741	January	Saturday	13	Low

Appendix B: PostgreSQL Database Implementation

This appendix documents the creation of the PostgreSQL database, the importation of the cleaned dataset, and the validation of the imported records.

Figure B1. PostgreSQL Database

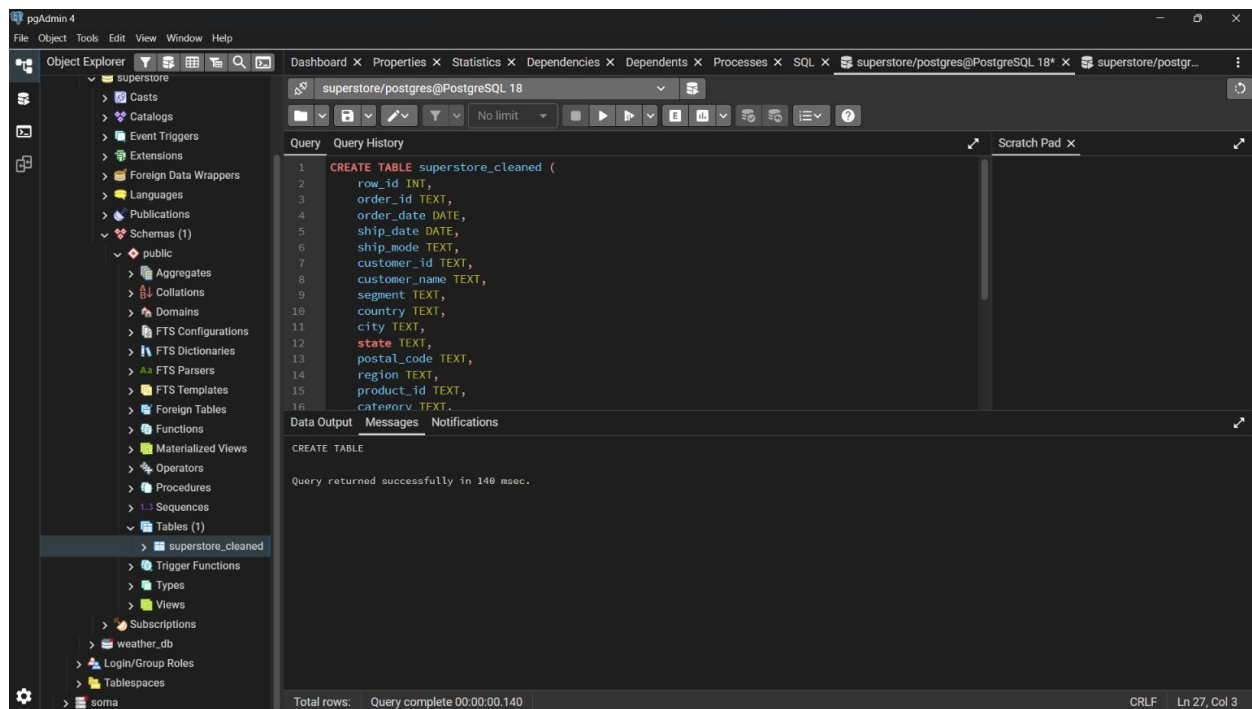


Figure B2. Database Table

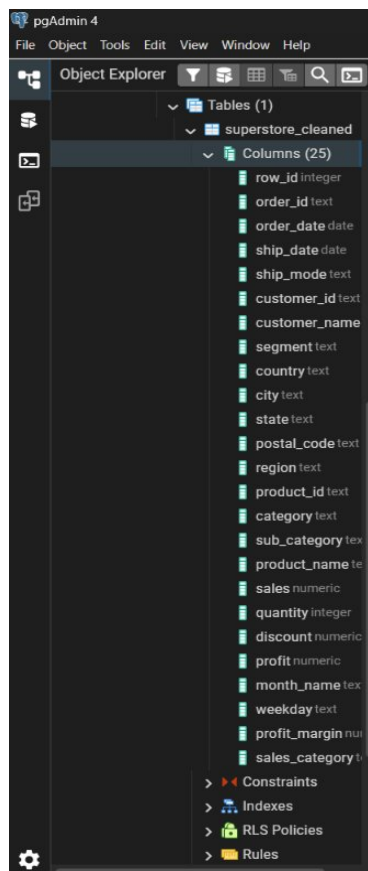


Figure B3. Successful Data Import

row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer
3	CA-2016-139608	2016-06-12	2016-06-16	Second Class	DV-13045	Darren Van Huff	Corporate
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer

Figure B4. Record Validation

count
9,994

Appendix C: SQL Queries

This appendix contains the SQL queries used during the exploratory data analysis phase to calculate key business metrics and answer analytical questions.

Figure C1. Total Sales Query

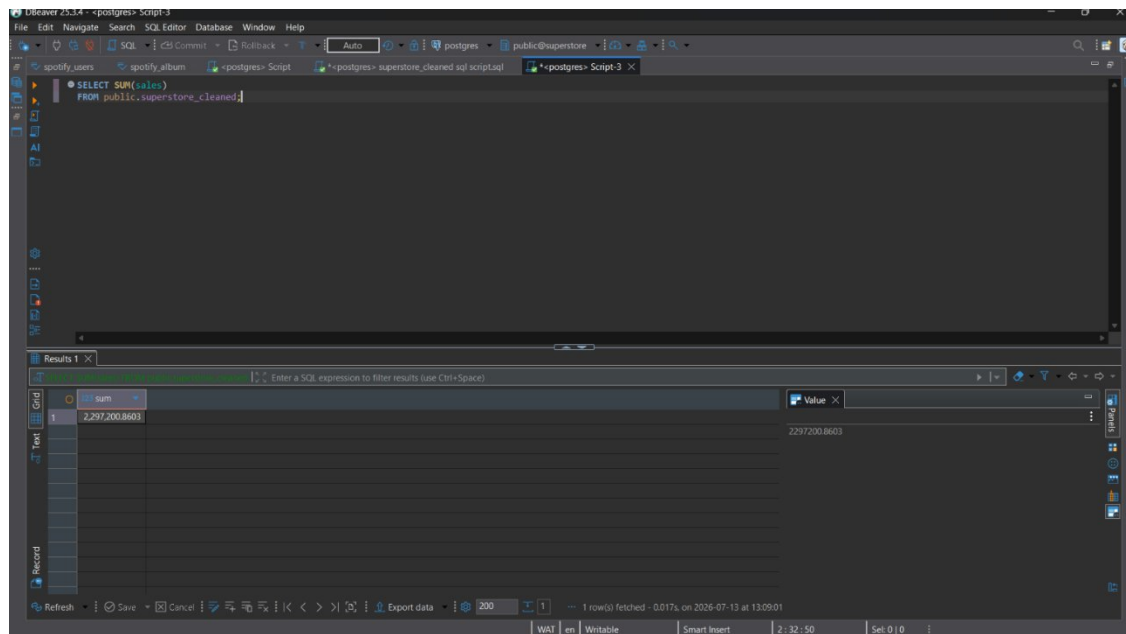


Figure C2. Total Profit Query

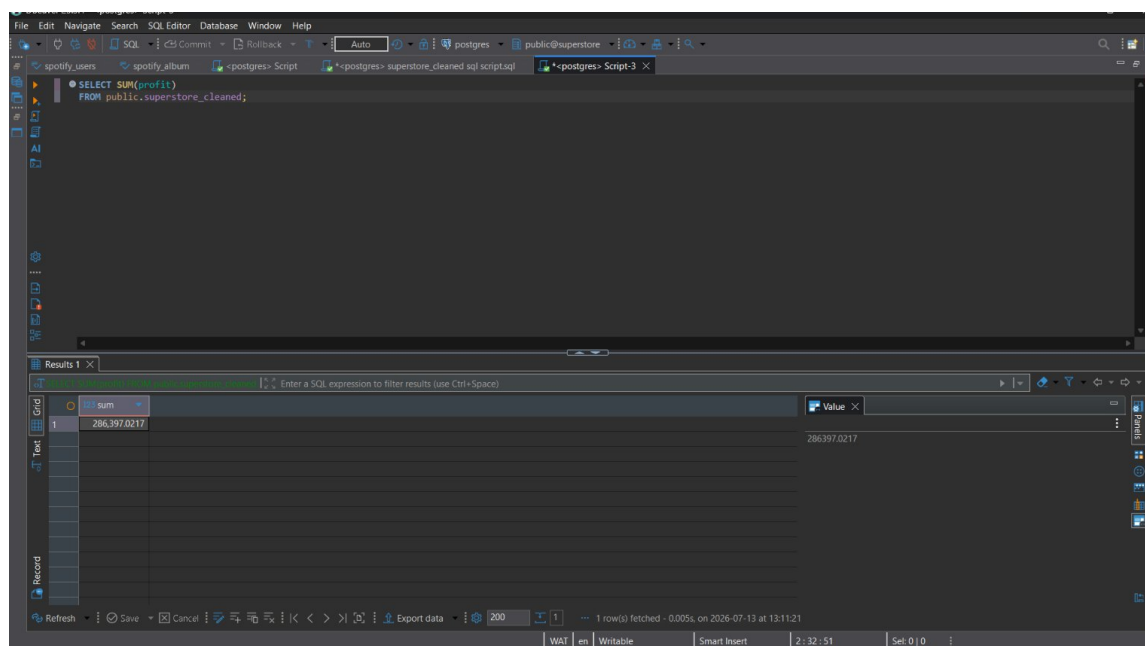


Figure C3. Sales by Category

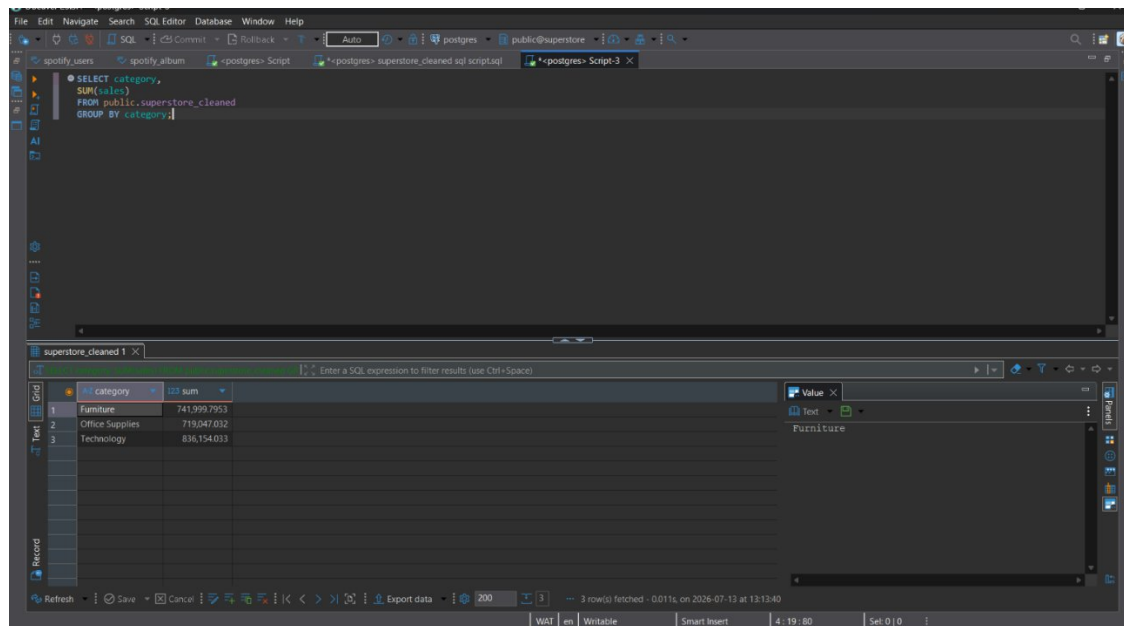


Figure C4. Profit by Region

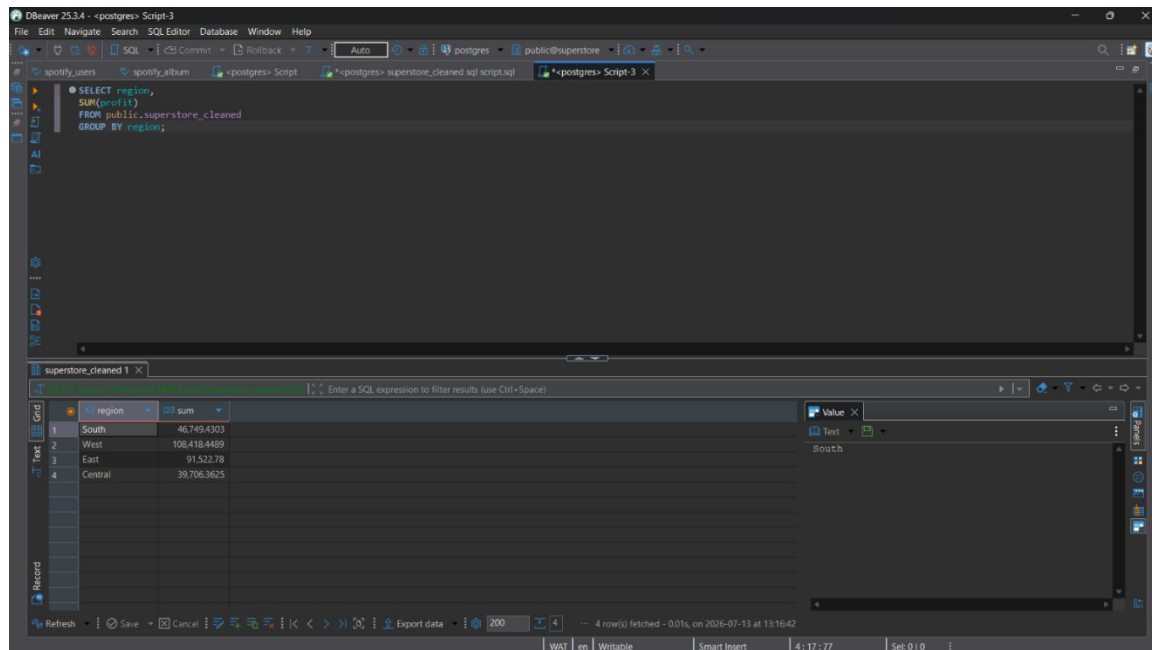


Figure C5. Sales by Customer Segment

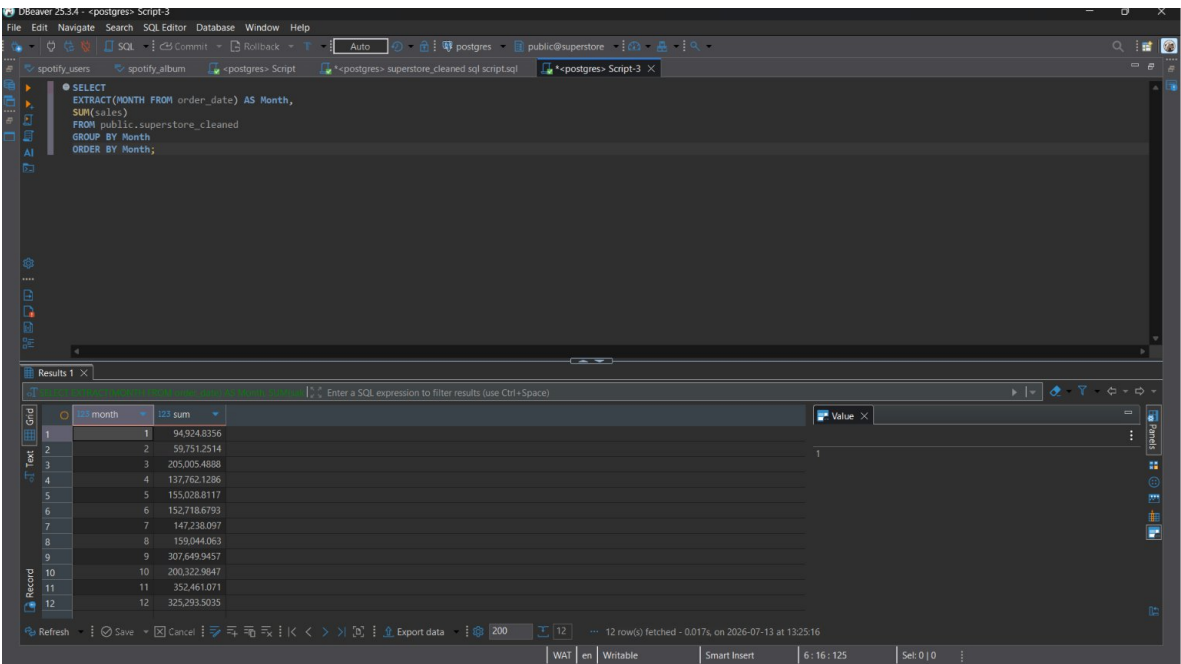
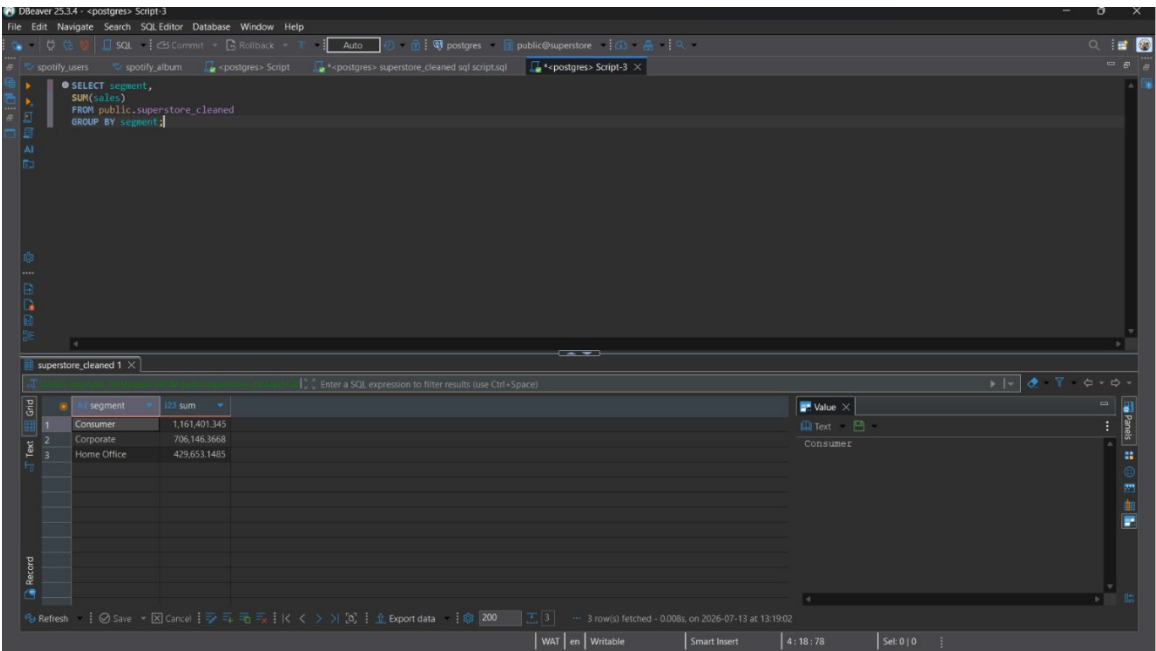


Figure C6. Monthly Sales Trend

Appendix D: DAX Measures

This appendix contains the DAX measures created in Microsoft Power BI to calculate the Key Performance Indicators (KPIs) displayed on the dashboard.

Figure D1. Total Sales Measure

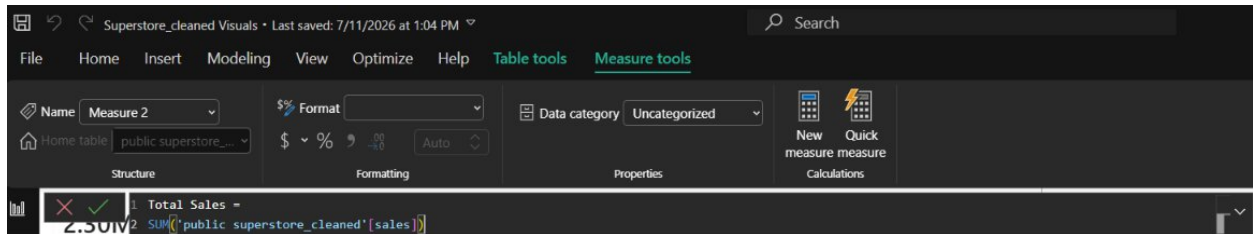


Figure D2. Total Profit Measure

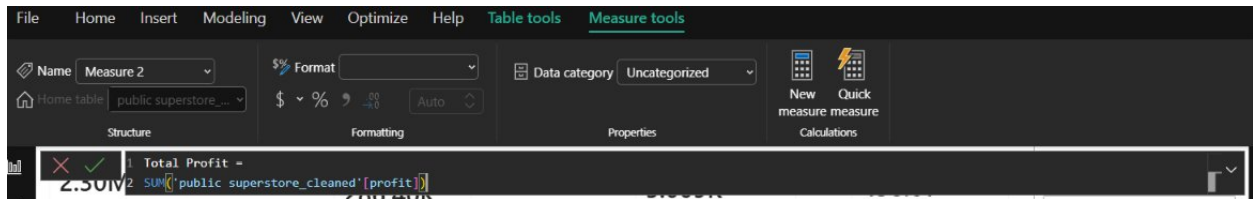


Figure D3. Total Orders Measure

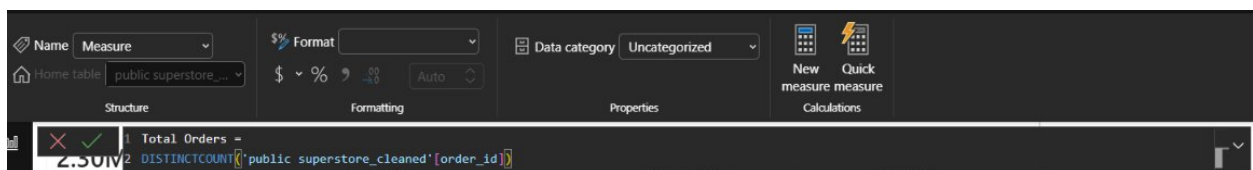
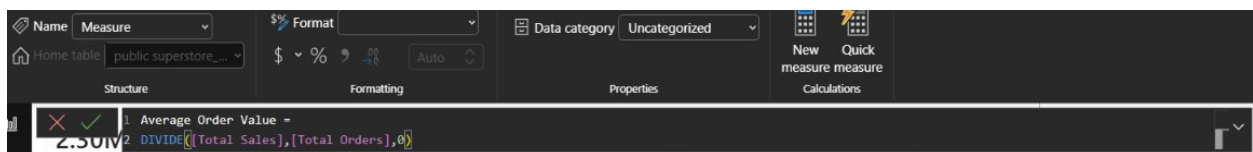


Figure D4. Average Order Value Measure



Appendix E: Power BI Dashboards

This appendix presents the completed dashboards developed during the project.

Figure E1. Executive Sales Dashboard

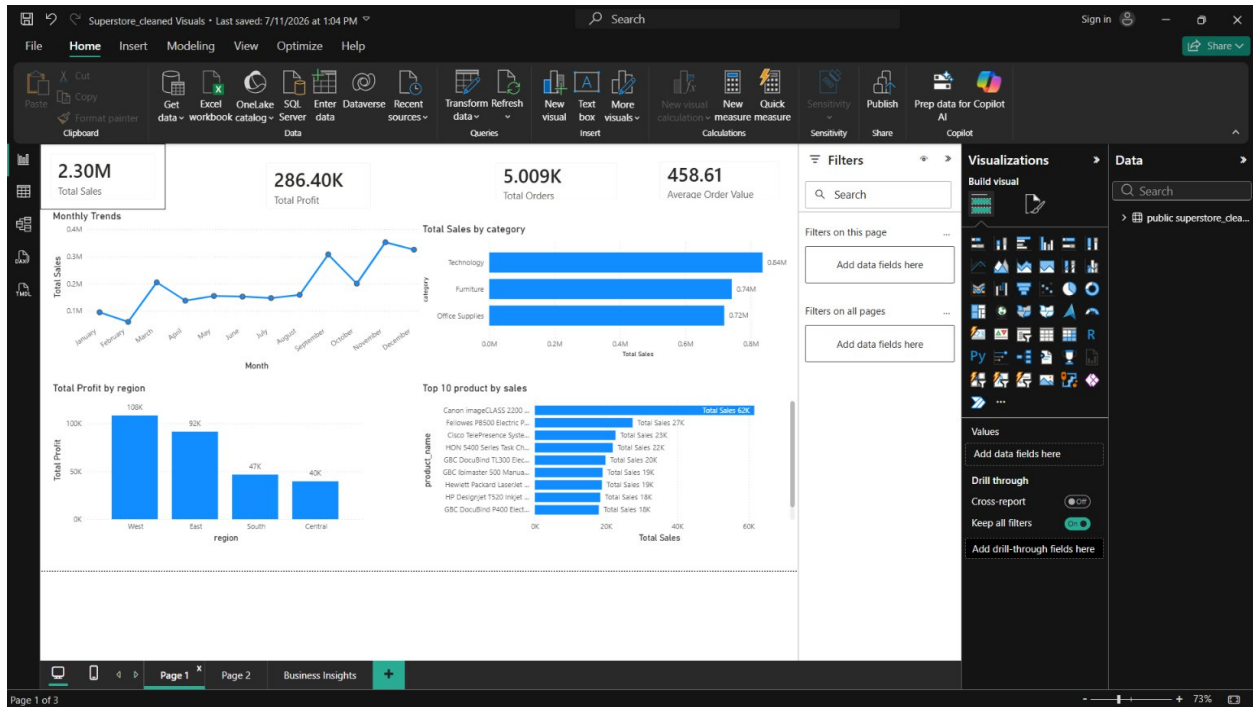


Figure E2. Interactive Dashboard

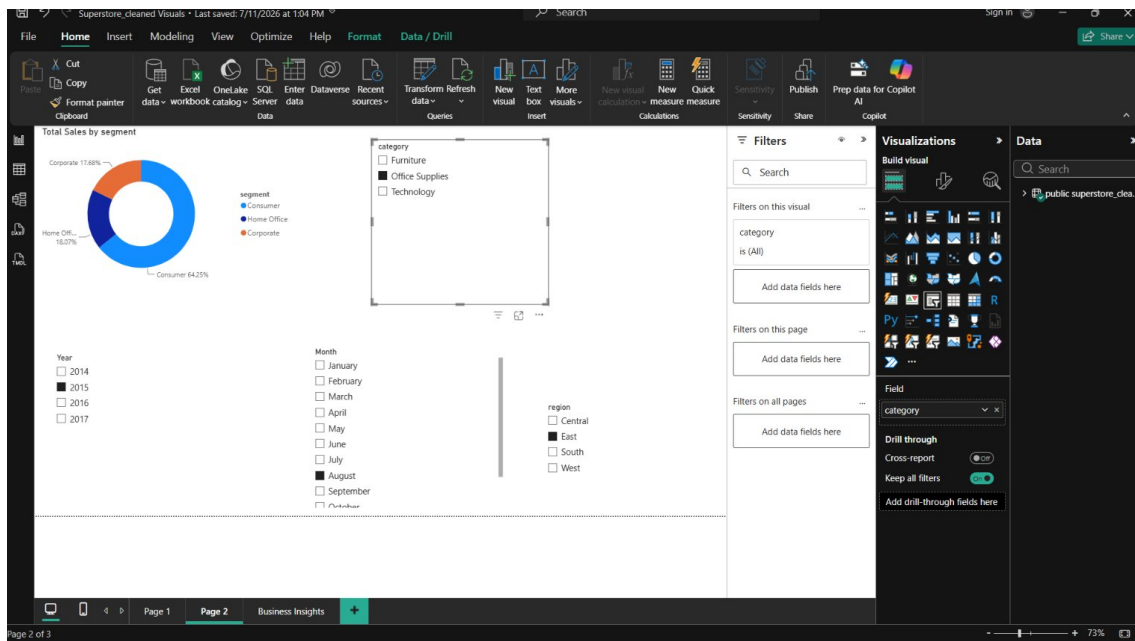


Figure E3. Business Insights Dashboard

